BrandLens

AI Brand Governance Platform - Round 2: The Build

Detail	Information
Team name	Double Trouble
Team leader / participant	Purvi Jain (solo team)
Live project	https://purvijain1234.github.io/BrandLens-AI-Brand-Governance-Platform/
Source repository (public)	https://github.com/Purvijain1234/BrandLens-AI-Brand-Governance-Platform
Problem statement	Inkloom AI - brand governance before publication

In one line: BrandLens lets a creator check a post or ad against approved brand rules before publishing, shows each issue with a reason and the rule it comes from, offers a fix the creator can accept or dismiss, and routes the content through reviewer comments and manager approval with a full history.

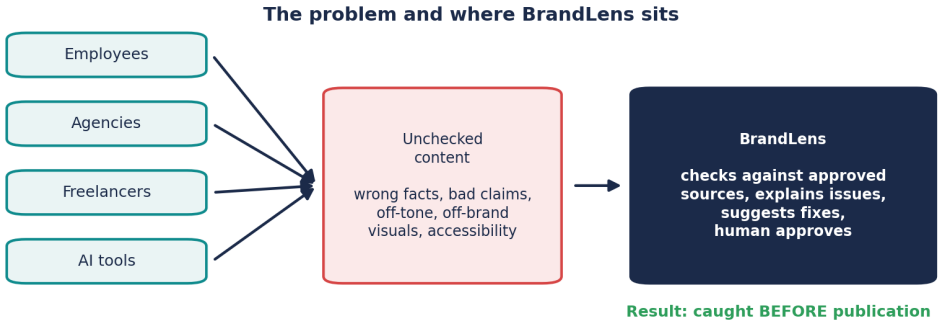


Figure 1: Where BrandLens sits between content creators and publication.

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1. Problem and Users

Brand content is now produced by employees, agencies, freelancers and AI tools. Each reads the guidelines differently, so content reaches the public with unsupported claims (for example "100% guaranteed results"), banned or inconsistent wording, outdated prices, a shouting tone, and accessibility or visual problems. These are found after publication, when they are hard to retract.

User	Need	What BrandLens gives them
Content creator	Find and fix problems before publishing	Highlighted findings with reasons, the source rule, and one-click suggested fixes
Reviewer	Make traceable decisions	Comments, change requests and the full history of what changed
Manager	Final sign-off with confidence	Approval that stays blocked while serious findings are unresolved
Brand admin	Keep rules current, spot unclear rules	Editable knowledge base and insights on rules that are often dismissed

2. Core Workflow (Working End to End)

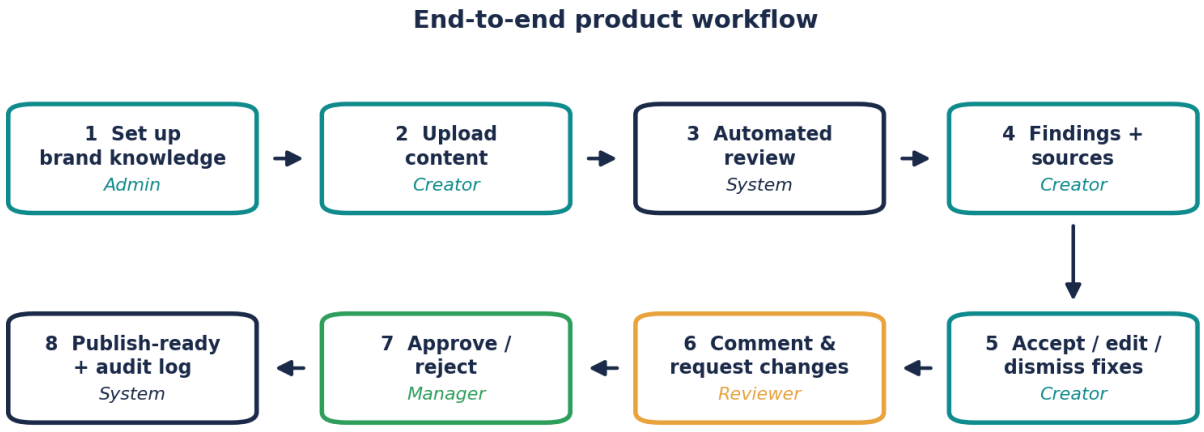


Figure 2: The workflow a judge can follow in the live project, from content to approval.

Step	What the user does	What the product does
1	Creator enters content and image settings (alt text, colours)	Stores a draft
2	Creator runs the brand review	Rule engine checks claims, terminology, price, tone, accessibility and palette
3	Creator reads the findings	Each shows severity, explanation, confidence and the guideline or fact it comes from
4	Creator accepts a fix or dismisses it with a reason	Accepted fixes edit the text; every decision is saved as a new version and logged
5	Creator submits for approval	Status changes to In review
6	Reviewer comments or requests changes; Manager approves	Approval is blocked while high-severity findings remain open

Step	What the user does	What the product does
7	Anyone checks history	Version history and audit log show who did what and when
8	Manager or admin opens Insights	Flag counts per clause and repeated dismissals highlight unclear guidelines

3. Completed Features

Only features that are built and usable in the live project are listed.

Area	Completed feature
Review	Interactive brand content review for social posts and ad copy
Claims	Detection of unsupported promotional claims and superlatives
Terminology	Approved-terminology checks with suggested replacements
Facts	Price validation against the approved price
Tone	Checks for excessive exclamation marks and ALL CAPS
Accessibility	Image alt-text check, colour-contrast check and readability check
Visual	Brand colour palette check on image settings
Explainability	Highlighted content with severity, explanation, confidence and guideline reference on each finding
Human control	Accept or dismiss suggested fixes; reason required when dismissing
Workflow	Creator, Reviewer, Manager and Admin roles; submit, comment, approve, request changes
Governance	Version history, audit log, editable terminology and approved price, insights dashboard
Transparency	AI disclosure note stating that reviews use rule-based checks and no external AI service

4. Technology and Architecture

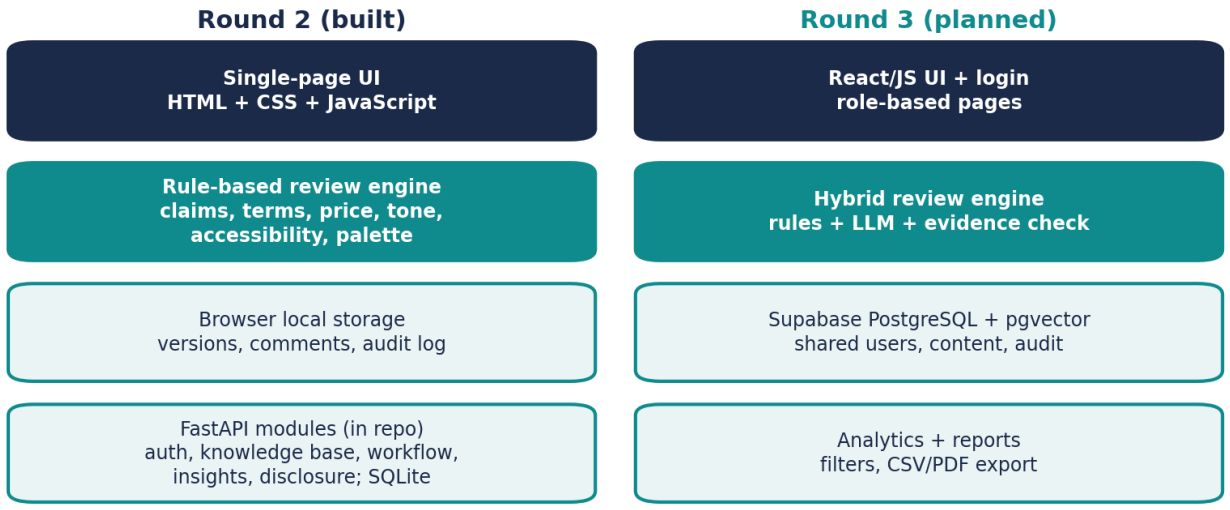


Figure 3: What is built in Round 2, and where Round 3 takes it.

Part	Technology	How it is used
Frontend	HTML, CSS (custom properties, dark mode), vanilla JavaScript	Single-page app with review, queue, knowledge base and insights views
Review engine	Rule-based analysis in JavaScript	Pattern and list checks for claims and terminology, price comparison, tone rules, WCAG-style contrast ratio, palette comparison, readability
Backend (in repository)	Python 3.13, FastAPI, Uvicorn, Pydantic, SQLite, JWT demo authentication	API modules for authentication, knowledge base, workflow, insights and AI disclosure
Hosting	GitHub Pages (frontend)	Public link that opens without sign-in
AI models / external APIs	None in the review path	Findings are deterministic and repeatable; no content leaves the browser

What was technically hard

- **Explainable, repeatable findings:** every finding is tied to a named clause, fact or term, so a reviewer can see exactly why it was raised.
- **Fix handling without breaking the text:** accepting a fix rewrites only the flagged span, then the content is re-reviewed so positions stay correct and earlier dismissals are remembered.
- **Human-control gate:** approval is enforced in code, not just in the interface; it stays blocked while high-severity findings are open.
- **Insight signal:** clauses that are dismissed in at least half of their flags are surfaced as possibly unclear guidelines.

5. Progress Since Round 1

Round 1 plan item	Round 2 status	Result
Brand knowledge base (terms, facts, palette)	Editable terminology, approved price and palette in the app	Done
Multi-category detection	Claims, terminology, facts, tone, accessibility, visual palette	Done
Explainable findings with sources	Severity, reason, confidence and clause reference on every finding	Done
Suggestions that never auto-override	Accept / dismiss with recorded reason	Done
Role-based review and approval	Four roles, comments, approve / request changes, gated approval	Done
Version history and audit log	Versions on every decision; audit log of actions	Done
Governance insights	Flags per clause and unclear-guideline signal	Partly (sample-history based, no filters yet)
Disclosure of AI use	Per-review note that only rule-based checks are used	Done
LLM judgement with citations and verification	Not in the live review path yet	Round 3

Round 1 plan item	Round 2 status	Result
Shared database, real accounts	FastAPI + SQLite modules exist; live app uses browser storage	Round 3
DOCX / PDF upload and OCR	Not built; text and image settings only	Round 3 stretch

6. What Changed from the Original Proposal, and Why

- **Replaced for now:** the LLM and retrieval review layer became a deterministic rule engine. Reason: a solo builder in a 48-hour window needed a core workflow that is fast, repeatable and fully explainable first. The google-genai dependency was removed so the review does not depend on an external service.
- **Replaced for now:** Supabase/PostgreSQL became browser storage in the live app, with a FastAPI + SQLite backend kept in the repository. Reason: GitHub Pages can only host static files, and this gave judges a link that opens instantly.
- **Added:** a gated approval rule, dismissal reasons, and a visible AI-disclosure note, because the problem statement stresses human control and transparency.
- **Deferred:** DOCX/PDF upload and OCR. They add little to the core workflow compared with the Round 3 improvements below.
- **Unchanged:** the idea, users, workflow and the principle that findings are advisory and humans decide.

7. What Does Not Work Yet

Stated openly so judges are not surprised:

- **Rule-based only.** The review finds what its rules cover. It does not understand context, so it can miss subtle issues or flag harmless text.
- **No LLM or retrieval in the live review.** Contextual rewrites and source-grounded AI findings are Round 3 work.
- **Data is stored in the browser.** The live app is not a shared workspace; content, comments and approvals are not visible to other users or other devices, and clearing browser data removes them.
- **Roles are a demo selector,** not real accounts or permissions.
- **Backend not connected to the live link.** The FastAPI modules run locally from the repository; the hosted page does not call them.
- **No file upload or OCR.** Content is entered as text with image settings.
- **Insights use sample starting history,** grow as reviews are done, and have no date filters or export yet.

8. Three Major Improvements for Round 3



Figure 4: The three improvements promised for Round 3.

Improvement 1: Evidence-grounded LLM review

What it is: a hybrid review engine. The existing rules stay as the fast, certain layer. A language model, behind a swappable provider interface, adds judgement for tone, messaging and claim support. Approved guidelines and facts are split into numbered clauses and retrieved for each part of the content; the model must return structured findings that quote the content and cite a retrieved clause. A verification step discards or downgrades any finding whose quote is not in the content or whose source was not retrieved, and marks it "needs human review".
What changes for the user: contextual issues that fixed rules cannot catch are found, suggested rewrites keep the creator's intent, and every AI finding shows its source. **How it will be shown:** a labelled test set of sample content with measured precision and recall.

Improvement 2: Persistent database and real user management

What it is: connect the app to a hosted PostgreSQL database with proper sign-up and login, replacing browser storage and the demo role selector. Roles (creator, reviewer, manager, admin) are enforced on the server, and content, versions, comments, approvals and audit history are stored per organisation.
What changes for the user: a team can collaborate across sessions and devices, a reviewer sees what a creator submitted, and approvals and history survive a browser refresh. **How it will be shown:** two accounts with different roles completing a full submit-to-approve cycle on separate devices.

Improvement 3: Governance analytics and reports

What it is: an analytics dashboard computed from real stored reviews: most frequent violations, most disputed clauses, correction acceptance rate, review turnaround time and trends by content type, with date filters and a downloadable report for brand managers.
What changes for the user: managers can see where mistakes repeat and which guidelines need rewriting, and can measure whether the review process is improving. **How it will be shown:** dashboard and exported report generated from reviews made in the live app, with no hard-coded numbers.

9. Responsible Design and Disclosure

Area	Approach
Human oversight	All findings are advisory. Fixes are never applied automatically, dismissals need a reason, and approval is blocked while serious findings are open.
Explainability	Each finding shows its reason, confidence and the clause, fact or term it comes from.

Area	Approach
Privacy	In Round 2 no content is sent to any external AI or server; review runs in the browser. Round 3 adds confidential-mode redaction before any model call.
Disclosure	The app states that reviews use rule-based checks only. AI coding assistants were used to help write and debug parts of the code; this is disclosed here and in the repository. No third-party templates were used for the core logic.
Limits	Findings may be wrong; manual review remains necessary (see Section 7).

10. Team and Contribution

Item	Details
Team name	Double Trouble
Team leader	Purvi Jain
Team members	Purvi Jain (solo participant)
Contribution	Product planning and research, UI design and frontend, rule-based review logic, FastAPI backend modules, architecture, testing, documentation and submission.